EDSTAQ ACADEMY MANAGEMENT PORTAL

Demand Analysis, Supply Modeling & Hiring Intelligence Specification

Complete Master Report: Inbound Demand Audit, Teacher Supply Cannibalization, and 4-Pillar Recruitment Engine

1. Allocation Feature Removal Documentation

As confirmed by academic operations, manual/new class allocation from the teacher table and timetable matrix was deemed redundant and has been completely decommissioned across the admin codebase:

- **Teachers Overview Table (`TeachersTable.jsx`):** Removed the 'Allocate' button from each teacher's row action cell. Only the Declines log modal remains accessible.
- **Schedule Matrix Modal (`ScheduleMatrixModal.jsx`):** Removed the 'New Allocation' header action button, preserving pure timetable and availability viewing.
- **Application Core (`App.jsx`):** Removed `TeacherAssignModal` imports, state references, allocation triggers, and local optimistic storage dependencies.

2. Lead & Demo Book: Inbound Student Demand Analysis

Analysis of **149 student demo records** against the Master Subjects catalog reveals current demand velocity and enrollment bottlenecks:

Demo Status	Count	Percentage	Strategic Significance
Converted to Sale	38	25.5%	Closed revenue admissions
Rejected	74	49.7%	Price sensitivity, schedule clashes, or decision-maker drop-offs
Scheduled (Upcoming)	18	12.1%	Active pipeline awaiting demo session
Completed	13	8.7%	Demo finished, pending final closing call
Follow up	6	4.0%	Interested leads scheduled for re-contact

2.1 Division / Level Demand Breakdown

- **Class 8:** #1 single grade in demand (**25 demos**, 35.3% conversion rate).
- **Class 4:** 18 demos, but 0 conversions due to price sensitivity.
- **Class 7:** 16 demos (25.0% conversion rate).
- **Class 3:** 16 demos (45.5% conversion rate).
- **Class 6:** 14 demos (33.3% conversion rate).
- **Class 9:** 12 demos (High dropout in Maths/Hindi).
- **Higher Secondary (HSS: Plus 1 & Plus 2):** 10 requests with an extraordinary **75.0% conversion rate**. High willingness to pay for commerce and science tuitions.

2.2 Core Subject Demand Breakdown

- **Mathematics:** 48 requests (32.2% of total demand). High volume across all grades.
- **English:** 39 requests (26.2% of total demand).
- **Hindi:** 22 requests (14.8% of total demand). Acute teacher shortage in CBSE Class 7 and Kerala Class 9.
- **Malayalam:** 13 requests (8.7% of total demand).

2.3 Qualitative Rejection Root-Cause Breakdown

> - **Pricing / Affordability (~45%):** Parents praise tutor quality, but struggle with 1-on-1 monthly fees. Multiple notes indicate demand for lower-priced group formats.
> - **Decision-Maker Gap (~25%):** Demos attended only by student/mother where financial decision-making lies with the father.
> - **School Exam / Calendar Clashes (~15%):** Temporary pauses due to USS examinations, Madrasa exam cycles, or school Kalolsavam.
> - **Alternative Preferences (~15%):** Offline home tuition enquiries or coaching centers.

3. The 4-Pillar Hiring Requirement Profile & Mathematical Model

As specified by Edstaq academic management, teacher recruitment must be qualified across four core parameters:

- **1. Target Subject:** Mathematics, English, Hindi, Basic Science, Physics, Chemistry, Accountancy.
- **2. Division / Stage:** Lower Primary (LP: 1-4), Upper Primary (UP: 5-7), High School (HS: 8-10), Higher Secondary (HSS: +1/+2).
- **3. Curriculum / Board:** Kerala State Syllabus vs. CBSE Syllabus (or Dual-Board competency).
- **4. Teaching Language / Medium:** English Medium, Malayalam Medium, or Bilingual / Mixed Delivery.

3.1 Mathematical Engine

```
H_required = max(0, ceil((Demand_hours - Supply_hours) / Standard_Capacity_per_Teacher))
If Demand >= 2 and Current_Active_Teachers = 0: H_required = max(1, ceil(Demand * 0.5))
```

3.2 Dynamic Reserve / Buffer Pool (B_reserve)

To absorb future trend spikes once Supply = Demand, a reserve buffer is maintained by level:

- **Higher Secondary (+1/+2):** 50% buffer (High conversion value ~75%)
- **High School (Classes 8-10):** 40% buffer (Board exam volume)
- **Upper Primary (Classes 5-7):** 30% buffer
- **Lower Primary (Classes 1-4):** 25% buffer

3.3 Urgent Priority Hiring Action Matrix

Priority	Subject & Curriculum Profile	Active Supply	Immediate Hires	Buffer Pool
🚨 P1 (Critical)	**Class 10 - MATHS CBSE (SUB318)** Board: CBSE | Language: English & Mixed	0 Teachers	**2 Hires**	+1 Reserve
🚨 P2 (Critical)	**Class 7 - HINDI CBSE (SUB239)** Board: CBSE | Language: English & Mixed	0 Teachers	**2 Hires**	+1 Reserve
🚨 P3 (High)	**Class 9 - HINDI Kerala (SUB296)** Board: Kerala | Language: Mixed / Malayalam	0 Teachers	**1 Hire**	+1 Reserve
⚠️ P4 (High)	**Class 8 - MATHS Kerala (SUB275 & SUB283)** Board: Kerala | Language: Malayalam & English	2 (Overbooked)	**3 Hires**	+1 Reserve
⚠️ P5 (High)	**Plus 1 - COMMERCE & SCIENCE (+1)** Board: Kerala/CBSE | Language: English	1 Teacher	**2 Hires**	+1 Reserve

4. The 'Phantom Supply' Dilemma & Cluster-Based Anti-Cannibalization

> **The Maneesha Case Study: Understanding Shared Resource Strain**
>
> Consider teacher Maneesha on the roster:
> • Maneesha is qualified to teach Mathematics for Classes 5 to 10 (6 classes).
> • She teaches both Kerala Syllabus and CBSE Syllabus (6 × 2 = 12 total subject combinations).
> • However, Maneesha has only 3 sessions of available time per week.
>
> If one student books Maneesha for 3 sessions/week in Class 10 CBSE Maths, Maneesha's capacity immediately drops to ZERO. Consequently, all other 11 combinations instantly lose their teacher supply!
>
> A naive system that looks at each subject individually creates a false sense of security (Phantom Supply). In reality, 1 teacher with 3 sessions can only service 1 student.

Session-Weighted Cluster Pooling Solution

- **Nominal Headcount vs True Available Session Capacity:** Distinguishing on-paper qualification from actual unbooked time slots in peak student hours (17:00-22:00).
- **Cross-Subject Cannibalization Factor (κ):** κ = Total_Qualified_Combinations / Available_Weekly_Sessions. When κ > 1.5, the resource is flagged as 'Hyper-Shared / High Strain'.
- **Competency Cluster Aggregation:** Instead of calculating Class 5, 6, 7 in silos, demand and supply are pooled by domain (e.g., 'High School Mathematics Cluster'). Hiring targets recommend versatile teachers capable of covering the cluster pool.
- **UI Strain Indicators:** Subject rows display a ' ⚠️ Shared Capacity (Stretched across N subjects)' badge to alert academic coordinators.

5. UI Implementation & Navigation Architecture

- **Navigation Switcher:** The top module dropdown ('Tutor Availability') in `Header.jsx` includes 'Supply vs Demand & Hiring Intel', switching views seamlessly.
- **KPI Summary Cards:** Active Demand, Teacher Pool, Critical Deficits, Immediate Hires Needed, and Reserve Pool.
- **4-Way Multi-Faceted Filter:** Division (LP/UP/HS/HSS), Board (Kerala/CBSE), Medium (English/Malayalam/Mixed), and Urgency Level.
- **Urgent Action Priority List:** Highlighted cards for immediate HR recruitment postings.
- **Interactive Matrix Table:** Full searchable breakdown with one-click 'Export HR Hiring Plan'.